

Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) **Write a C code to create a process using fork() system call.**
- b) **Read the user input as shell command.**
- c) **Use a system call exec() to execute the shell command.**
- d) **Use a wait() system call for waiting a parent process.**
- e) **Display the PIDs of parent and child processes.**

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("Fork failed");
        return 1;
    }

    if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());
        printf("Executing command: %s\n", command);

        execlp(command, command, (char *)NULL);

        perror("Execution failed");
        exit(1);
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID : %d\n", getpid());

```

```

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
|

```

```
ramya@LAPTOP-OPDAF309: ~$ cd ~
ramya@LAPTOP-OPDAF309: ~$ mkdir -p Practical2
ramya@LAPTOP-OPDAF309: ~$ cd Practical2
ramya@LAPTOP-OPDAF309: ~/Practical2$ nano task1.c
ramya@LAPTOP-OPDAF309: ~/Practical2$ gcc task1.c -o task1
ramya@LAPTOP-OPDAF309: ~/Practical2$ ./task1
Enter a Linux command: ls
Parent Process
Parent PID : 537
Child Process
Child PID : 538
Parent PID : 537
Executing command: ls
task1 task1.c
Child process completed.
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

```
ramya@LAPTOP-OPDAF309: ~$ uname -a
Linux LAPTOP-OPDAF309 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC
Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
ramya@LAPTOP-OPDAF309: ~$ lscpu
Architecture:                x86_64
  CPU op-mode(s):            32-bit, 64-bit
  Address sizes:              39 bits physical, 48 bits virtual
  Byte Order:                 Little Endian
CPU(s):                       12
  On-line CPU(s) list:       0-11
Vendor ID:                    GenuineIntel
  Model name:                 13th Gen Intel(R) Core(TM) i5-13420H
    CPU family:                6
    Model:                     186
    Thread(s) per core:        2
    Core(s) per socket:        6
    Socket(s):                 1
    Stepping:                  2
    BogoMIPS:                  5222.40
  Flags:                      fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities
Virtualization features:
  Virtualization:             VT-x
  Hypervisor vendor:           Microsoft
  Virtualization type:         full
Caches (sum of all):
  L1d:                         288 KiB (6 instances)
  L1i:                         192 KiB (6 instances)
```

```
ramya@LAPTOP-OPDAF309: ~$ cat /proc/meminfo
L2: 7.5 MiB (6 instances)
L3: 12 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-11
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Old microcode: Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DISABLE
Srbds: Not affected
Tsa: Not affected
Tsx async abort: Not affected
Vmscape: Not affected
ramya@LAPTOP-OPDAF309: ~/Practical2$ ps
  PID TTY          TIME CMD
  318 pts/0    00:00:00 bash
  615 pts/0    00:00:00 ps
ramya@LAPTOP-OPDAF309: ~/Practical2$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.2   0.1  24124 15420 ?        Ss   16:01   0:01 /sbin/init
root         2   0.0   0.0    3180   2212 hvc0     Sl+  16:01   0:00 /init
root         7   0.0   0.0    3180   2048 hvc0     Sl+  16:01   0:00 plan9 --c
root        50   0.0   0.2  50404 16644 ?        S<s  16:01   0:00 /usr/lib/
systemd+   85   0.0   0.1  22416 14480 ?        Ss   16:01   0:00 /usr/lib/
root        92   0.1   0.1  35248 12344 ?        Ss   16:01   0:00 /usr/lib/
```

```

root          92  0.1  0.1  35248 12344 ?          Ss   16:01  0:00 /usr/lib/
root         114  0.0  0.0   2888  1952 ?          Ss   16:01  0:00 /bin/sh /
root         115  0.0  0.0   4472  3160 ?          Ss   16:01  0:00 /usr/sbin
message+     116  0.0  0.0   8808  5368 ?          Ss   16:01  0:00 @dbus-dae
root         120  0.0  0.3  44096 29856 ?          Ss   16:01  0:00 /usr/bin/
root         165  0.0  0.1  18440  9392 ?          Ss   16:01  0:00 /usr/lib/
root         175  0.0  0.1 1866032 15536 ?        Ssl  16:01  0:00 /usr/libe
syslog       189  0.0  0.0 220548  5908 ?        Ssl  16:01  0:00 /usr/sbin
root        215  0.0  0.4 123460 32780 ?        Ssl  16:01  0:00 /usr/bin/
_chrony      233  0.0  0.0  20424  6468 ?          S    16:01  0:00 /usr/sbin
_chrony      238  0.0  0.0  12096  2424 ?          S    16:01  0:00 /usr/sbin
root        240  0.0  0.0   5492  2752 tty1       Ss+  16:01  0:00 /usr/sbin
root        313  0.0  0.0   3184  1116 ?          Ss   16:01  0:00 /init
root        314  0.0  0.0   3200  1252 ?          S    16:01  0:00 /init
ramya       318  0.0  0.0   6268  5536 pts/0      Ss   16:01  0:00 -bash
root        320  0.0  0.0   8644  5576 ?          Ss   16:01  0:00 login --
ramya       384  0.0  0.1  22344 13560 ?          Ss   16:01  0:00 /usr/lib/
ramya       386  0.0  0.0  22920  4116 ?          S    16:01  0:00 (sd-pam)
ramya       414  0.0  0.0   6136  5480 pts/1      Ss+  16:01  0:00 -bash
root        620  0.0  0.0  35252  6464 ?          S    16:09  0:00 (udev-wor
root        621  0.0  0.0  35252  6464 ?          S    16:09  0:00 (udev-wor
ramya       627  0.0  0.0   7160  4336 pts/0      R+   16:09  0:00 ps aux

```

ramya@LAPTOP-OPDA5309:~/Practical2\$ top

```

ramya@LAPTOP-OPDAF309:~/Practical2$ top
top - 16:12:26 up 10 min, 1 user, load average: 0.08, 0.04, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 7756.6 total, 7213.0 free, 535.8 used, 158.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7220.9 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	24124	15420	11624	S	0.0	0.2	0:01.29
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.02
7	root	20	0	3180	2048	1940	S	0.0	0.0	0:00.00
50	root	19	-1	50404	16652	15388	S	0.0	0.2	0:00.27
85	systemd+	20	0	22416	14480	11984	S	0.0	0.2	0:00.10
92	root	20	0	35248	12344	9248	S	0.0	0.2	0:00.65
114	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.06
115	root	20	0	4472	3160	2908	S	0.0	0.0	0:00.01
116	message+	20	0	8808	5368	4680	S	0.0	0.1	0:00.08
120	root	20	0	44096	29856	14716	S	0.0	0.4	0:00.36
165	root	20	0	18440	9392	8144	S	0.0	0.1	0:00.11
175	root	20	0	1866032	13800	10808	S	0.0	0.2	0:00.13
189	syslog	20	0	220548	5908	4604	S	0.0	0.1	0:00.08
215	root	20	0	123460	32780	18172	S	0.0	0.4	0:00.29
233	_chrony	20	0	20424	6468	5688	S	0.0	0.1	0:00.05
238	_chrony	20	0	12096	2424	1780	S	0.0	0.0	0:00.00
240	root	20	0	5492	2752	2564	S	0.0	0.0	0:00.01
313	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00
314	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.07
318	ramya	20	0	6268	5536	3864	S	0.0	0.1	0:00.14
320	root	20	0	8644	5576	4736	S	0.0	0.1	0:00.03
384	ramya	20	0	22344	13560	10952	S	0.0	0.2	0:00.29
386	ramya	20	0	22920	4116	2112	S	0.0	0.1	0:00.00
414	ramya	20	0	6136	5480	3928	S	0.0	0.1	0:00.08
637	ramya	20	0	8172	6000	3888	R	0.0	0.1	0:00.12